

Piecewise Self-Similar Radiation-Hydrodynamics Analytic Profiles

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1 Wave Front and Mass Coordinates

All physical quantities are scaled using nanoseconds (ns) for time, g/cm^2 for mass coordinates, g/cm^3 for density, MBar for pressure, km/s for velocity, and HeV for temperature.

Exact physical constants (according to SI):

Quantity	value	SI units
c	299792458	m / s
k_B	1.380649×10^{-23}	J / K
q	$1.602176634 \times 10^{-19}$	J
h	$6.62607015 \times 10^{-34}$	J s

Other fundamental quantities derived are:

Quantity	Theoretical Expression (SI)	CGS Units
HeV	$\frac{q}{k_B} \cdot 100$	K
σ_{SB}	$\frac{2\pi^5 k_B^4}{15c^2 h^3}$	$\text{erg}/(\text{cm}^2 \cdot \text{s} \cdot \text{K}^4)$
a	$\frac{4\sigma_{SB}}{c}$	$\text{erg}/(\text{cm}^3 \cdot \text{K}^4)$

Case 1: $\tau = 0$

Material model.

Quantity	value	CGS units
f	3.4×10^{13}	erg / g
f_{Kelvin}	$\frac{f}{\text{HeV}^\beta}$	erg / g
β	1.6	-
μ	0.14	
g	7200	cm^2 / g
g_{Kelvin}	$\frac{g}{\text{HeV}^\alpha}$	erg / g
α	1.5	-
λ	0.2	-
γ	1.25	-
r	0.25	-
ρ_0	19.32	g/cm^3

1.1 Self Similar Constants

$$A = T_0(rf)^{\frac{1}{\beta}} = 120472196.47695 \quad (1)$$

$$B = \frac{16\sigma g}{3\beta(rf)^{\frac{4+\alpha}{\beta}}} = 1.70920e - 28 \quad (2)$$

$$n = \frac{4 + \alpha}{\beta} = \frac{55}{16} \quad (3)$$

$$k = 1 - \mu = 1 - 0.14 = 0.86 \quad (4)$$

$$a_1 = \frac{\mu}{\beta} = \frac{7}{80} \quad (5)$$

$$a_2 = \frac{2 - \mu}{\beta} = \frac{93}{80} \quad (6)$$

$$a_3 = -\frac{2}{\beta} - \tau = -\frac{5}{4} \quad (7)$$

$$b_1 = \frac{\beta(2 + \lambda) - \mu(4 + \alpha)}{\beta} = \frac{55}{32} \quad (8)$$

$$b_2 = \frac{2\beta - \beta\lambda - 8 - 2\alpha + \mu(4 + \alpha)}{\beta} = -\frac{723}{160} \quad (9)$$

$$b_3 = \frac{8 - 3\beta + 2\alpha}{\beta} = \frac{31}{8} \quad (10)$$

$$a = \frac{2\beta - \beta\lambda - 8 - 2\alpha + \mu(4 + \alpha)}{4 + 2\lambda - 4\mu} = -\frac{245}{128} \quad (11)$$

$$b = \frac{\mu - 2}{4 + 2\lambda - 4\mu} = -\frac{31}{64} \quad (12)$$

$$c = \frac{3\mu - 2\lambda - 2 + \tau(2(\beta - \alpha - 4) - \beta\lambda + \mu(4 + \alpha))}{4 + 2\lambda - 4\mu} = -\frac{33}{64} \quad (13)$$

$$a_v = \frac{2\beta - 4 - \alpha}{2 + \lambda - 2\mu} = -\frac{115}{96} \quad (14)$$

$$b_v = \frac{-1}{2 + \lambda - 2\mu} = -\frac{25}{48} \quad (15)$$

$$c_v = \frac{1 - \tau(4 + \alpha - 2\beta)}{2 + \lambda - 2\mu} = \frac{25}{48} \quad (16)$$

$$a_u = \frac{\beta(2 + \lambda) - \mu(4 + \alpha)}{4 + 2\lambda - 4\mu} = \frac{275}{384} \quad (17)$$

$$b_u = \frac{-\mu}{4 + 2\lambda - 4\mu} = -\frac{7}{192} \quad (18)$$

$$c_u = \frac{\tau(\beta(2+\lambda) - \mu(4+\alpha)) + \mu}{4+2\lambda-4\mu} = \frac{7}{192} \quad (19)$$

$$a_p = \frac{\beta\lambda + 4 + \alpha - \mu(4+\alpha)}{2+\lambda-2\mu} = \frac{505}{192} \quad (20)$$

$$b_p = \frac{1-\mu}{2+\lambda-2\mu} = \frac{43}{96} \quad (21)$$

$$c_p = \frac{\tau(\beta\lambda + 4 + \alpha - \mu(4+\alpha)) + \mu - 1}{2+\lambda-2\mu} = -\frac{43}{96} \quad (22)$$

1.2 Wave Coordinate Normalization Variables

The ablated mass coordinate $m_f(t)$ and shocked mass coordinate $m_s(t)$ grow as exact power laws in time:

$$m_f(t) = \xi_f A^{-a} B^{-b} t^{-c} \approx 0.00102 \left(\frac{t}{\text{ns}} \right)^{\frac{33}{64}} \text{ g/cm}^2 \quad (23)$$

$$m_s(t) = m_{s0} \left(\frac{t}{\text{ns}} \right)^{\frac{149}{192}} \approx 0.01209 \left(\frac{t}{\text{ns}} \right)^{\frac{149}{192}} \text{ g/cm}^2 \quad (24)$$

So we can define the normalized mass coordinates $y = m/m_f(t)$ and $y_s = m/m_s(t)$ directly as:

$$y = \frac{m}{m_f(t)} \approx 981.45247 m \left(\frac{t}{\text{ns}} \right)^{-\frac{33}{64}} \quad (25)$$

$$y_s = \frac{m}{m_s(t)} \approx 82.68924 m \left(\frac{t}{\text{ns}} \right)^{-\frac{149}{192}} \quad (26)$$

2 Explicit Subsonic Ablation Profiles ($m < m_f(t)$)

Fluid Velocity $u(m, t)$

1. Dimensionless Fit Function:

$$U(y) \approx -6.76974 \frac{1-y}{1+5.34241y}$$

2. Level 1 (Theoretical Scaling):

$$u(m, t) = U(y) A^{a_u} B^{b_u} t^{c_u} = U(y) A^{\frac{275}{384}} B^{-\frac{7}{192}} t^{\frac{7}{192}}$$

3. Level 2 (Parametrized Shape):

$$u(m, t) \approx -6.76974 \frac{1-y}{1+5.34241y} A^{\frac{275}{384}} B^{-\frac{7}{192}} t^{\frac{7}{192}}$$

4. **Level 2.5 (Substitute Constants at 1 ns):** Using physical conversions ($u_{\text{scale}} \approx 29.60156 \text{ km/s}$):

$$\begin{aligned} u(m, t) &\approx -6.76974 \frac{1-y}{1+5.34241y} \cdot 29.60156 \left(\frac{t}{\text{ns}} \right)^{\frac{7}{192}} \text{ km/s} \\ &\approx -200.39472 \frac{1-y}{1+5.34241y} \left(\frac{t}{\text{ns}} \right)^{\frac{7}{192}} \text{ km/s} \end{aligned}$$

5. **Level 3 (Fully Substituted m, t function):** Substituting $y \approx 981.45247 m \left(\frac{t}{\text{ns}} \right)^{-\frac{33}{64}}$:

$$u(m, t) \approx -200.39472 \frac{1 - 981.45247 m \left(\frac{t}{\text{ns}} \right)^{-\frac{33}{64}}}{1 + 5243.32440 m \left(\frac{t}{\text{ns}} \right)^{-\frac{33}{64}}} \left(\frac{t}{\text{ns}} \right)^{\frac{7}{192}} \text{ km/s}$$

Specific Volume $v(m, t)$

1. **Dimensionless Fit Function:**

$$V(y) \approx 483.36492 (1-y)^{39.54884}$$

2. **Level 1 (Theoretical Scaling):**

$$v(m, t) = V(y) A^{a_v} B^{b_v} t^{c_v} = V(y) A^{-\frac{115}{96}} B^{-\frac{25}{48}} t^{\frac{25}{48}}$$

3. **Level 2 (Parametrized Shape):**

$$v(m, t) \approx 483.36492 (1-y)^{39.54884} A^{-\frac{115}{96}} B^{-\frac{25}{48}} t^{\frac{25}{48}}$$

4. **Level 2.5 (Substitute Constants at 1 ns):** Using physical conversions ($v_{\text{scale}} \approx 1.24268 \text{ cm}^3/\text{g}$):

$$\begin{aligned} v(m, t) &\approx 483.36492 (1-y)^{39.54884} \cdot 1.24268 \left(\frac{t}{\text{ns}} \right)^{\frac{25}{48}} \text{ cm}^3/\text{g} \\ &\approx 600.66640 (1-y)^{39.54884} \left(\frac{t}{\text{ns}} \right)^{\frac{25}{48}} \text{ cm}^3/\text{g} \end{aligned}$$

5. **Level 3 (Fully Substituted m, t function):** Substituting $y \approx 981.45247 m \left(\frac{t}{\text{ns}} \right)^{-\frac{33}{64}}$:

$$v(m, t) \approx 600.66640 \left(1 - 981.45247 m \left(\frac{t}{\text{ns}} \right)^{-\frac{33}{64}} \right)^{39.54884} \left(\frac{t}{\text{ns}} \right)^{\frac{25}{48}} \text{ cm}^3/\text{g}$$

Pressure $p(m, t)$

1. **Dimensionless Fit Function:**

$$P(y) \approx 0.34846 y^{0.87633} + 0.02822 y^{19.97029}$$

2. **Level 1 (Theoretical Scaling):**

$$p(m, t) = P(y) A^{a_p} B^{b_p} t^{c_p} = P(y) A^{\frac{505}{192}} B^{\frac{43}{96}} t^{-\frac{43}{96}}$$

3. **Level 2 (Parametrized Shape):**

$$p(m, t) \approx [0.34846 y^{0.87633} + 0.02822 y^{19.97029}] A_{192}^{505} B_{96}^{43} t^{-\frac{43}{96}}$$

4. **Level 2.5 (Substitute Constants at 1 ns):** Using physical conversions ($p_{\text{scale}} \approx 7.05133$ MBar):

$$\begin{aligned} p(m, t) &\approx 7.05133 [0.34846 y^{0.87633} + 0.02822 y^{19.97029}] \left(\frac{t}{\text{ns}} \right)^{-\frac{43}{96}} \text{ MBar} \\ &\approx [2.45709 y^{0.87633} + 0.19901 y^{19.97029}] \left(\frac{t}{\text{ns}} \right)^{-\frac{43}{96}} \text{ MBar} \end{aligned}$$

5. **Level 3 (Fully Substituted m, t function):** Substituting $y \approx 981.45247 m \left(\frac{t}{\text{ns}} \right)^{-\frac{33}{64}}$:

$$p(m, t) \approx \left[2.45709 \left(981.45247 m \left(\frac{t}{\text{ns}} \right)^{-\frac{33}{64}} \right)^{0.87633} + 0.19901 \left(981.45247 m \left(\frac{t}{\text{ns}} \right)^{-\frac{33}{64}} \right)^{19.97029} \right] \left(\frac{t}{\text{ns}} \right)^{-\frac{43}{96}}$$

Temperature $T(m, t)$

1. **Dimensionless Fit Function:**

$$T(y) \approx [(1 - y)(1 + 0.20287y)]^{\frac{10}{39}}$$

2. **Level 1 (Theoretical Scaling):**

$$T(m, t) = T(y) T_0$$

3. **Level 2 (Parametrized Shape):**

$$T(m, t) \approx [(1 - y)(1 + 0.20287y)]^{\frac{10}{39}} T_0$$

4. **Level 2.5 (Substitute Constants at 1 ns):** Using physical boundary drive ($T_0 = 1.00000$ eV):

$$T(m, t) \approx 1.00000 [(1 - y)(1 + 0.20287y)]^{\frac{10}{39}} \text{ eV}$$

5. **Level 3 (Fully Substituted m, t function):** Substituting $y \approx 981.45247 m \left(\frac{t}{\text{ns}} \right)^{-\frac{33}{64}}$:

$$T(m, t) \approx 1.00000 \left[\left(1 - 981.45247 m \left(\frac{t}{\text{ns}} \right)^{-\frac{33}{64}} \right) \left(1 + 199.10933 m \left(\frac{t}{\text{ns}} \right)^{-\frac{33}{64}} \right) \right]^{\frac{10}{39}} \text{ eV}$$

Density $\rho(m, t)$

Note: The subsonic density profile diverges sharply at the ablation front ($\xi \rightarrow 1$), so instead of independently fitting $\tilde{\rho}(\xi)$, we compute it explicitly using the equation of state (EOS) along with the well-behaved analytical fits for pressure $p(m, t)$ and temperature $T(m, t)$.

1. **Level 1 (Theoretical relation):**

$$\rho(m, t) = \left(\frac{r f T(m, t)^\beta}{p(m, t)} \right)^{\frac{1}{\mu-1}}$$

2. **Level 2 (CGS Parametrized form):**

$$\rho(m, t) = \left(\frac{r f_{\text{Kelvin}} T(m, t)^\beta}{p(m, t)} \right)^{\frac{1}{\mu-1}}$$

3. **Level 3 (Explicit numerical relation in g/cm³):** Using $r = 0.25$, $f_{\text{Kelvin}} = 3.4 \times 10^{13}/(\text{KELVIN_PER_HEV})^{1.6}$, $\beta = 1.6$, and $\mu = 0.14$:

$$\rho(m, t) = \left(\frac{r f_{\text{Kelvin}} T(m, t)^{1.6}}{p(m, t)} \right)^{-\frac{1}{0.86}} \text{ g/cm}^3$$

3 Explicit Shock compressed Profiles ($m_f(t) < m < m_s(t)$)

Fluid Velocity $u(m, t)$

1. **Dimensionless Fit Function:**

$$U(y_s) \approx 0.97536 + (0.17755) y_s^{0.63757}$$

2. **Level 1 (Theoretical Scaling):**

$$u(m, t) = U(y_s) v_0^{\frac{1}{2}} p_0^{\frac{1}{2}} t^{\frac{\tau}{2}}$$

3. **Level 2 (Parametrized Shape):**

$$u(m, t) \approx [0.97536 + (0.17755) y_s^{0.63757}] v_0^{\frac{1}{2}} p_0^{\frac{1}{2}} t^{\frac{\tau}{2}}$$

4. **Level 2.5 (Substitute Constants at 1 ns):** Using physical conversions ($u_{\text{scale}} \approx 3.74525 \text{ km/s}$):

$$\begin{aligned} u(m, t) &\approx [0.97536 + (0.17755) y_s^{0.63757}] \cdot 3.74525 \left(\frac{t}{\text{ns}} \right)^{-\frac{43}{192}} \text{ km/s} \\ &\approx [3.65295 + 0.66499 y_s^{0.63757}] \left(\frac{t}{\text{ns}} \right)^{-\frac{43}{192}} \text{ km/s} \end{aligned}$$

5. **Level 3 (Fully Substituted m, t function):** Substituting $y_s \approx 82.68924 m \left(\frac{t}{\text{ns}} \right)^{-\frac{149}{192}}$:

$$u(m, t) \approx \left[3.65295 + 11.10002 m^{0.63757} \left(\frac{t}{\text{ns}} \right)^{-0.49478} \right] \left(\frac{t}{\text{ns}} \right)^{-\frac{43}{192}} \text{ km/s}$$

Pressure $p(m, t)$

1. **Dimensionless Fit Function:**

$$P(y_s) \approx 1.0 + 0.49535 y_s^{1.13315}$$

2. **Level 1 (Theoretical Scaling):**

$$p(m, t) = P(y_s) p_0 t^\tau$$

3. **Level 2 (Parametrized Shape):**

$$p(m, t) \approx [1.0 + 0.49535 y_s^{1.13315}] p_0 t^\tau$$

4. **Level 2.5 (Substitute Constants at 1 ns):** Using physical boundary drive ($p_0 = 2.71000$ MBar):

$$\begin{aligned} p(m, t) &\approx 2.71000 [1.0 + 0.49535 y_s^{1.13315}] \left(\frac{t}{\text{ns}} \right)^{-\frac{43}{96}} \text{ MBar} \\ &\approx [2.71000 + 1.34240 y_s^{1.13315}] \left(\frac{t}{\text{ns}} \right)^{-\frac{43}{96}} \text{ MBar} \end{aligned}$$

5. **Level 3 (Fully Substituted m, t function):** Substituting $y_s \approx 82.68924 m \left(\frac{t}{\text{ns}} \right)^{-\frac{149}{192}}$:

$$p(m, t) \approx \left[2.71000 + 199.81724 m^{1.13315} \left(\frac{t}{\text{ns}} \right)^{-0.87937} \right] \left(\frac{t}{\text{ns}} \right)^{-\frac{43}{96}} \text{ MBar}$$

Density $\rho(m, t)$

1. **Dimensionless Fit Function:**

$$V(y_s)^{-1} \approx 9.00000 y_s^{0.64939}$$

2. **Level 1 (Theoretical Scaling):**

$$\rho(m, t) = \frac{1}{v_0} V(y_s)^{-1}$$

3. **Level 2 (Parametrized Shape):**

$$\rho(m, t) \approx \rho_0 \rho_s y_s^{0.64939}$$

4. **Level 2.5 (Substitute Constants at 1 ns):** Using physical density $\rho_0 = 19.32$ g/cm³:

$$\rho(m, t) \approx 173.88000 y_s^{0.64939} \text{ g/cm}^3$$

5. **Level 3 (Fully Substituted m, t function):** Substituting $y_s \approx 82.68924 m \left(\frac{t}{\text{ns}} \right)^{-\frac{149}{192}}$:

$$\rho(m, t) \approx 3057.82924 m^{0.64939} \left(\frac{t}{\text{ns}} \right)^{-0.50395} \text{ g/cm}^3$$

Temperature $T(m, t)$

In the shocked region, the temperature profile $T(m, t)$ is derived algebraically from the equation of state (EOS) along with the well-behaved fits for pressure $p(m, t)$ and density $\rho(m, t)$:

1. Level 1 (Theoretical relation):

$$T(m, t) = \left(\frac{p(m, t) v(m, t)^{1-\mu}}{r f} \right)^{\frac{1}{\beta}} = \left(\frac{p(m, t) \rho(m, t)^{\mu-1}}{r f} \right)^{\frac{1}{\beta}}$$

2. Level 2 (CGS Parametrized form):

$$T(m, t) \approx \left(\frac{P(y_s) p_0 t^\tau (\rho_0 \rho_s y_s^{d_p})^{\mu-1}}{r f_{\text{Kelvin}}} \right)^{\frac{1}{\beta}}$$

3. Level 2.25 (Algebraic simplification of bracket):

We pull the power $y_s^{(\mu-1)}$ out of the bracket raised to $1/\beta = 0.625$, grouping all constants:

$$T(y_s, t) = T_{\text{scale}} \cdot y_s^{-0.7125 d_p} \left[1.0 + (P_s - 1.0) y_s^{d_p} \right]^{0.625} t^{-\frac{215}{768}}$$

4. Level 2.5 (Substitute Constants at 1 ns):

Using evaluated prefactors ($T_{\text{scale.hev}} \approx 0.01240 \text{ HeV}$):

$$T(m, t) \approx 0.01240 \cdot y_s^{-0.46269} \left[1.0 + 0.49535 y_s^{1.13315} \right]^{0.625} \left(\frac{t}{\text{ns}} \right)^{-\frac{215}{768}} \text{ HeV}$$

Or in Kelvin: $T(m, t) \approx 14393.59038 \cdot y_s^{-0.46269} \left[1.0 + 0.49535 y_s^{1.13315} \right]^{0.625} \left(\frac{t}{\text{ns}} \right)^{-\frac{215}{768}} \text{ Kelvin}$

5. Level 3 (Fully Substituted m, t function in HeV):

Substituting $y_s \approx 82.68924 m \left(\frac{t}{\text{ns}} \right)^{-\frac{149}{192}}$:

$$T(m, t) \approx 0.00161 \left[m^{-0.74030} \left(\frac{t}{\text{ns}} \right)^{0.57450} + 73.73330 m^{0.39285} \left(\frac{t}{\text{ns}} \right)^{-0.30487} \right]^{0.625} \left(\frac{t}{\text{ns}} \right)^{-\frac{215}{768}} \text{ HeV}$$